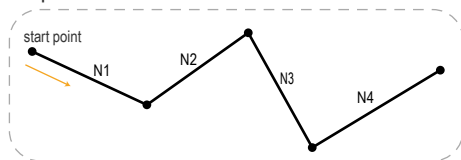
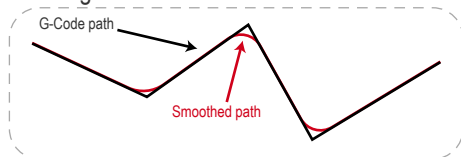


a) Traditional Serial Planning

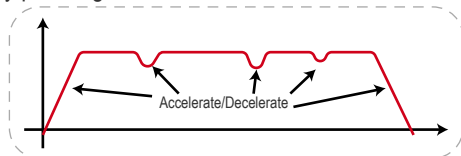
a.1 G-Code Input



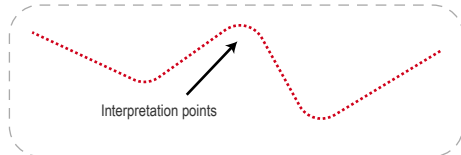
a.2 Path smoothing



a.3 Velocity planning



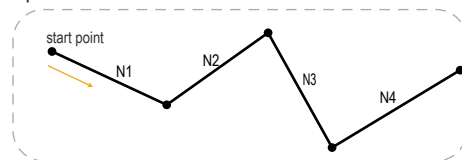
a.4 Interpretation



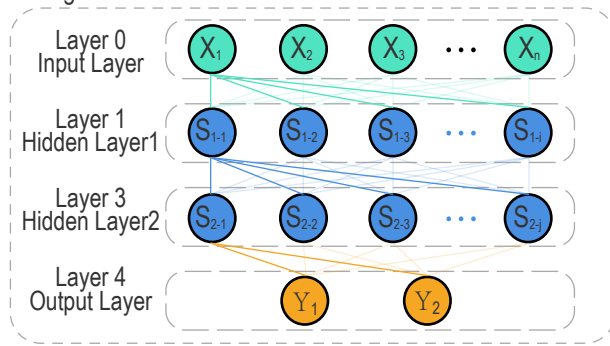
a.5 Servo command

b) J-NNC end-to-end Planning

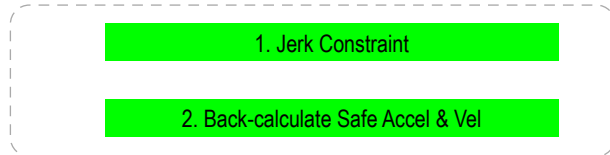
b.1 G-Code Input



b.2 PPO Agent



b.3 Kinematic Constraint Module



b.4 Safe Servo Command